National Qualifications 2019

X807/75/02

Biology
Section 1 — Questions

TUESDAY, 30 APRIL

1:00 PM – 3:30 PM

Instructions for the completion of Section 1 are given on *page 02* of your question and answer booklet X807/75/01.

Record your answers on the answer grid on *page 03* of your question and answer booklet.

Before leaving the examination room you must give your question and answer booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

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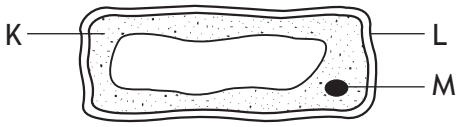
SECTION 1 — 25 marks

Attempt ALL questions

1. Animal cells left in a solution with a lower water concentration than their contents

A shrink
B burst
C become turgid
D become plasmolysed.

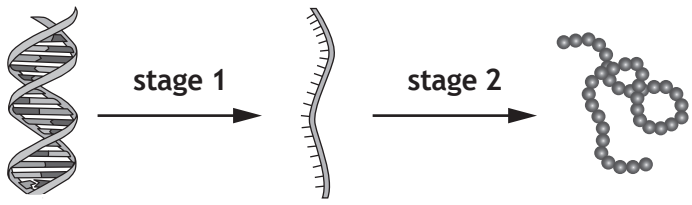
2. The diagram represents a typical plant cell.



Which of the labelled parts could also be found in a typical fungal cell?

A L and M
B K and M
C K and L
D K, L and M

3. The diagram shows stages in the production of a protein in a cell.



Which row in the table identifies the exact location of each stage?

	Stage 1	Stage 2
A	nucleus	cytoplasm
B	nucleus	ribosome
C	cytoplasm	ribosome
D	cytoplasm	nucleus

4. A single strand of DNA contains 830 adenine, 929 cytosine, 774 guanine and 615 thymine bases.

How many guanine bases would be in the complementary strand?

A 615

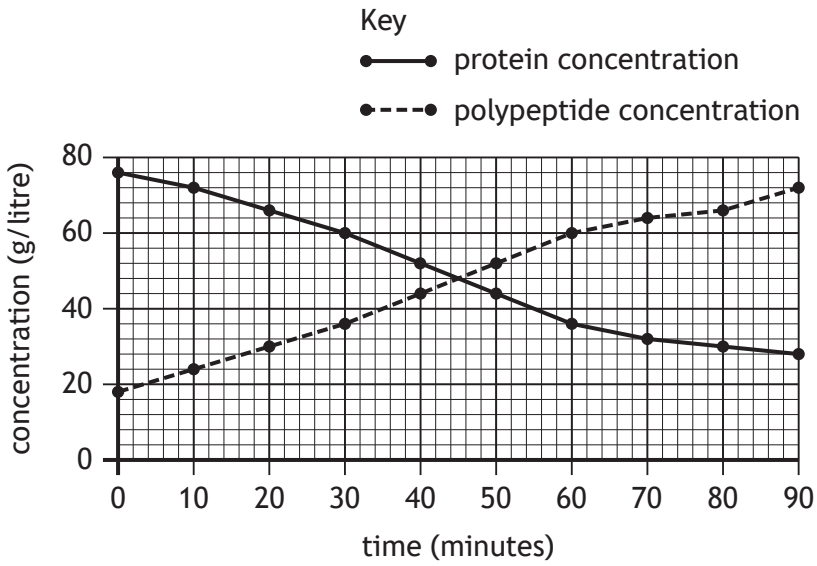
B 774

C 830

D 929

5. Proteins are broken down in the stomach into polypeptides.

The graph shows the concentration of proteins and polypeptides in the stomach over 90 minutes.



The ratio of protein concentration to polypeptide concentration in the stomach after 30 minutes is

A 3:5

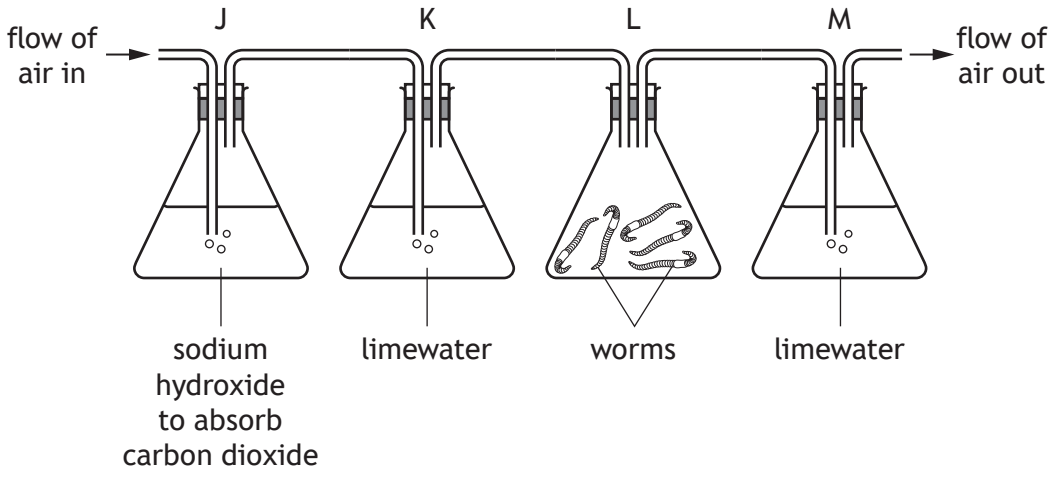
B 5:3

C 13:7

D 7:13

[Turn over

6. Four flasks, J, K, L and M, were set up to investigate the production of carbon dioxide during respiration.



Limewater turns increasingly cloudy as more carbon dioxide is passed through it.

Predict what would happen if only one worm was used in flask L.

The limewater in flask

A K would turn cloudy more slowly

B K would turn cloudy more quickly

C M would turn cloudy more slowly

D M would turn cloudy more quickly.

7. Which of the following reactions takes place during fermentation in plant cells?

A pyruvate ⟶ carbon dioxide + ethanol

B glucose ⟶ carbon dioxide + water

C glucose ⟶ water + ethanol

D pyruvate ⟶ lactate

8. A cell with 10 chromosomes divided by mitosis.

Which row in the table identifies the number of daughter cells produced and the number of chromosomes in each daughter cell?

	Number of daughter cells produced	**Number of chromosomes in each daughter cell**
A	1	20
B	1	5
C	2	10
D	2	5

9. Which row in the table shows the type of message that is transferred through various structures in a reflex arc?

	Sensory neuron	**Synapse**	**Motor neuron**
A	chemical	electrical	chemical
B	electrical	chemical	electrical
C	chemical	chemical	electrical
D	electrical	electrical	chemical

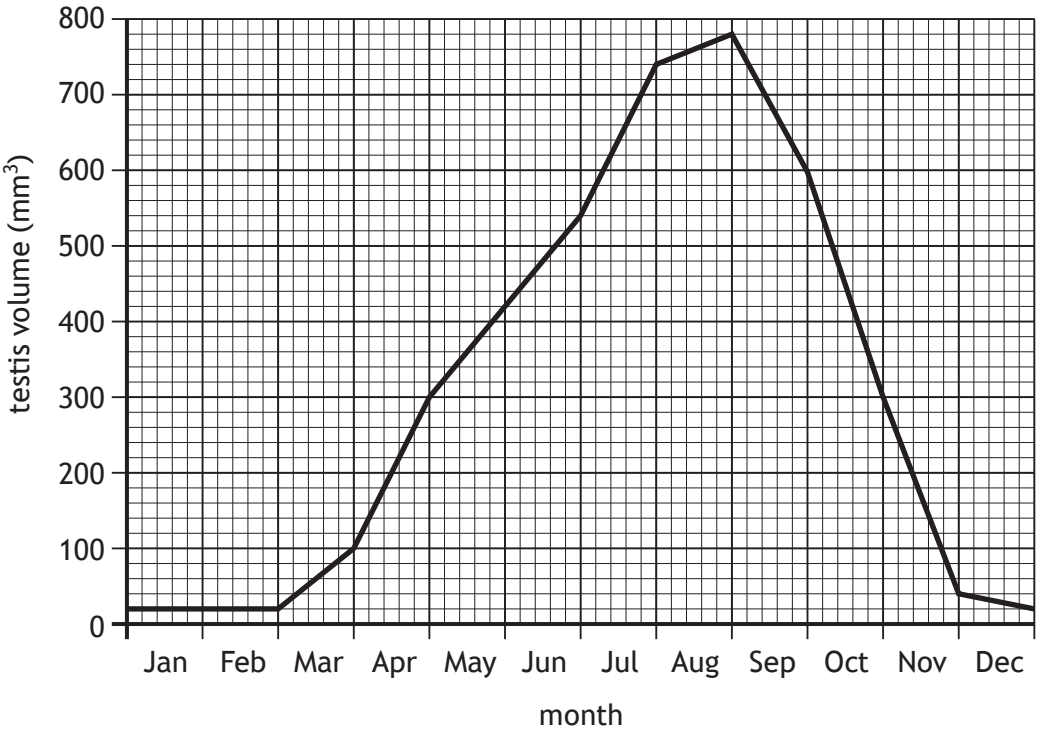
10. Hormones are released by

A endocrine glands
B blood cells
C receptor cells
D target tissues.

[Turn over

11. The volume of one bird's testis was measured on the last day of each month for a year.

The graph shows the results.



Which of the following statements is true?

The volume of the testis

A is constant from end of November to end of February

B increases more between end of March and end of April than any other month

C increases for only five months of the year

D decreases for only four months of the year.

12. Which term describes the type of variation in which a characteristic is controlled by more than one gene?

A Continuous

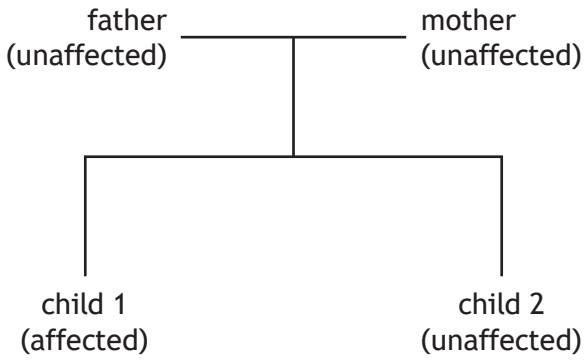
B Discrete

C Polygenic

D Heterozygous

13. Albinism is a condition in which the production of a pigment that colours the skin is limited. It is controlled by a recessive allele.

The diagram shows how a family was affected by the condition.



The chance of this couple's 3rd child being affected by the condition is

A 1 in 1

B 1 in 2

C 1 in 3

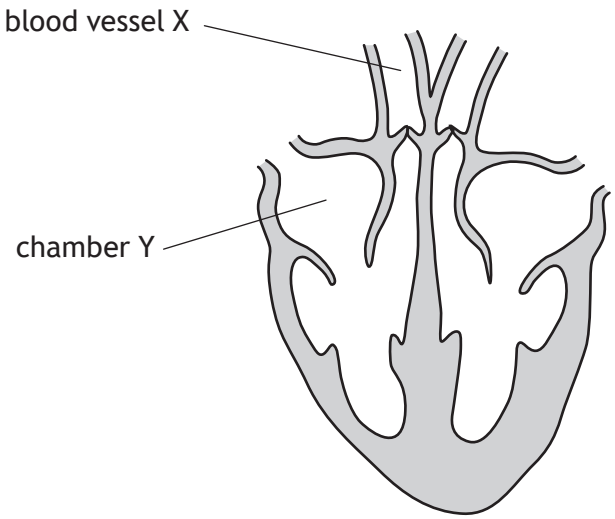
D 1 in 4.

14. Which row in the table describes features of phloem?

	Substance transported	**Sieve plates**
A	sugar	present
B	sugar	absent
C	water	absent
D	water	present

[Turn over

15. The diagram shows a mammalian heart and associated blood vessels.



Which row in the table identifies blood vessel X and chamber Y?

	Blood vessel X	**Chamber Y**
A	pulmonary artery	left atrium
B	pulmonary vein	right atrium
C	pulmonary artery	right atrium
D	pulmonary vein	left atrium

16. The table shows the composition of some of the gases in inhaled and exhaled air.

Gas	**Gas composition (%)**	
	Inhaled air	**Exhaled air**
Oxygen	20	16
Carbon dioxide	0·04	4

How many times greater is the carbon dioxide concentration in exhaled air than in inhaled air?

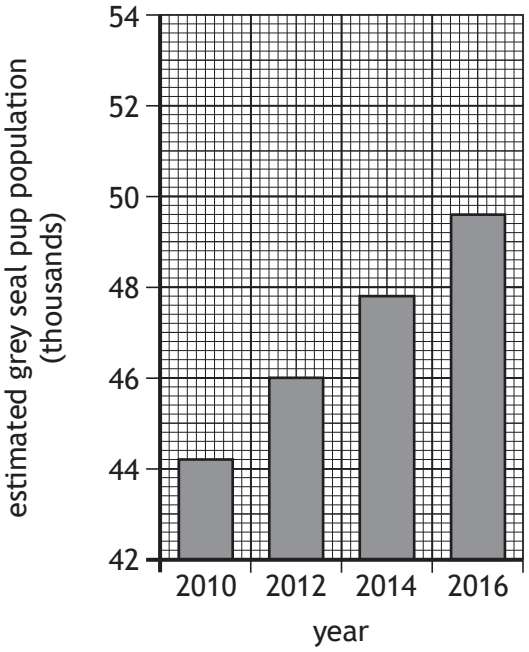
A 0·16

B 3·96

C 100

D 500

17. The graph shows survey results of estimated grey seal pup populations every two years from 2010 to 2016.



If the grey seal pup population continues to increase by the same number at each survey, what will the estimated population be in 2020?

A 51 400

B 53 200

C 55 000

D 56 800

18. Competition occurs when required resources are in short supply.

Interspecific competition occurs when individuals of

A the same species compete for a few of the same resources

B different species compete for a few of the same resources

C the same species compete for all the same resources

D different species compete for all the same resources.

[Turn over

19. At six different sample sites in a stream, the oxygen concentration, pH and numbers of different organisms were recorded.

The higher the number of organisms in the sample the more abundant they are.

The results are shown in the tables.

Table 1

Factor	Sample site					
	1	**2**	**3**	**4**	**5**	**6**
Oxygen concentration (units)	88	80	75	72	28	61
pH	5·7	6·0	6·6	7·3	7·6	8·0

Table 2

Organism	Sample site					
	1	**2**	**3**	**4**	**5**	**6**
Mayfly nymphs	0	0	0	5	7	122
Dragonfly nymphs	3	3	2	3	2	2
Chironimid fly larvae	0	1	1	2	227	32
Freshwater snails	0	0	0	0	50	75

Using the results from both tables identify which of the following conclusions is **false**.

A Freshwater snails do not survive in water with a lower pH

B Changes in pH have little effect on the distribution of dragonfly nymphs

C Mayfly nymphs are at their most abundant when the oxygen concentration is lowest

D Chironimid fly larvae are at their most abundant when the oxygen concentration is lowest

20. In which parts of a green leaf would most photosynthesis occur?

A Palisade mesophyll and lower epidermis

B Lower epidermis and guard cells

C Guard cells and spongy mesophyll

D Spongy mesophyll and palisade mesophyll

21. The table shows the rate of photosynthesis in a plant under different light intensities.

Light intensity (kilolux)	Rate of photosynthesis (units)
10	2
20	28
30	53
40	76
50	85

Which change in light intensity produces the greatest increase in the rate of photosynthesis?

A 10 to 20 kilolux
B 20 to 30 kilolux
C 30 to 40 kilolux
D 40 to 50 kilolux

22. A gardener decided to treat his crops with both fertiliser and pesticides.
The result of this would be

A a decrease in soil nitrates and an increase in crop yield
B an increase in soil nitrates and an increase in crop yield
C an increase in soil nitrates and a decrease in crop yield
D a decrease in soil nitrates and a decrease in crop yield.

23. Which of the following could occur as a result of fertiliser leaching into a fresh water pond?

	Algae population	Bacterial population	Oxygen concentration
A	increases	increases	increases
B	decreases	decreases	decreases
C	decreases	decreases	increases
D	increases	increases	decreases

[Turn over for next question

24. For the successful biological control of whitefly in a greenhouse, it was recommended to use 50 individuals of a predator species to kill a population of 1500 whitefly.

The number of predators that would be required to kill 21 000 whitefly is

A 420

B 700

C 19 500

D 75 000.

25. Which of the following statements describe the possible effects of a mutation on the survival of an organism?

1. It has no effect
2. It gives the organism an advantage
3. It disadvantages the organism

A 1 and 2 only

B 1 and 3 only

C 2 and 3 only

D 1, 2 and 3

[END OF SECTION 1. NOW ATTEMPT THE QUESTIONS IN SECTION 2 OF YOUR QUESTION AND ANSWER BOOKLET.]

FOR OFFICIAL USE

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National Qualifications 2019

Mark

X807/75/01

Biology
Section 1 — Answer grid and Section 2

TUESDAY, 30 APRIL
1:00 PM – 3:30 PM

X807750 1

Fill in these boxes and read what is printed below.

Full name of centre

Town

Forename(s)

Surname

Number of seat

Date of birth

Day | Month | Year

Scottish candidate number

Total marks — 100

SECTION 1 — 25 marks

Attempt ALL questions.

Instructions for completion of Section 1 are given on *page 02*.

SECTION 2 — 75 marks

Attempt ALL questions.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers and rough work is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting. Any rough work must be written in this booklet. Score through your rough work when you have written your final copy.

Use **blue** or **black** ink.

Before leaving the examination room you must give this booklet to the Invigilator; if you do not, you may lose all the marks for this paper.

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X807750101

SECTION 1 — 25 marks

The questions for Section 1 are contained in the question paper X807/75/02.

Read these and record your answers on the answer grid on *page 03* opposite.

Use **blue** or **black** ink. Do NOT use gel pens or pencil.

1. The answer to each question is **either** A, B, C or D. Decide what your answer is, then fill in the appropriate bubble (see sample question below).

2. There is **only one correct** answer to each question.

3. Any rough working should be done on the additional space for answers and rough work at the end of this booklet.

Sample question

The thigh bone is called the

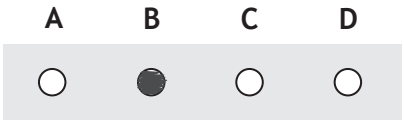
A humerus

B femur

C tibia

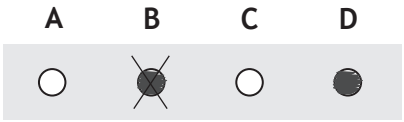
D fibula.

The correct answer is **B** — femur. The answer **B** bubble has been clearly filled in (see below).

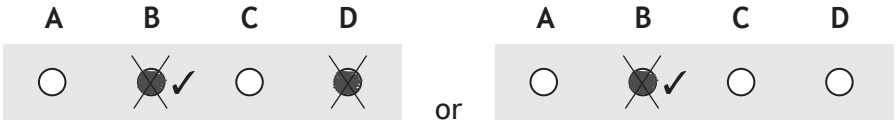


Changing an answer

If you decide to change your answer, cancel your first answer by putting a cross through it (see below) and fill in the answer you want. The answer below has been changed to **D**.



If you then decide to change back to an answer you have already scored out, put a tick (✓) to the **right** of the answer you want, as shown below:



SECTION 1 — Answer grid

0BJ25AD1

	A	B	C	D
1	○	○	○	○
2	○	○	○	○
3	○	○	○	○
4	○	○	○	○
5	○	○	○	○
6	○	○	○	○
7	○	○	○	○
8	○	○	○	○
9	○	○	○	○
10	○	○	○	○
11	○	○	○	○
12	○	○	○	○
13	○	○	○	○
14	○	○	○	○
15	○	○	○	○
16	○	○	○	○
17	○	○	○	○
18	○	○	○	○
19	○	○	○	○
20	○	○	○	○
21	○	○	○	○
22	○	○	○	○
23	○	○	○	○
24	○	○	○	○
25	○	○	○	○

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SECTION 2 — 75 marks

Attempt ALL questions

1. (a) The list gives four types of cells.

Bacteria Fungus Animal Plant

(i) Cell membranes are found in all of these cell types.

Describe the function of the cell membrane. **1**

_____

_____

(ii) Name **one** other structure that is also present in all of these cells. **1**

_____

(b) The table gives information on the numbers of mitochondria in different types of mammalian cells.

Cell type	Number of mitochondria per cell			
	Cell 1	Cell 2	Cell 3	Average
Muscle	1352	1203	1450	1335
Skin epithelium	250	330	275	
Lymphocyte	953	1112	860	975

(i) Complete the table by calculating the average number of mitochondria per cell in skin epithelium. **1**

Space for calculation

(ii) Compared to skin epithelium cells, muscle and lymphocyte cells have higher numbers of mitochondria.

Suggest why these cells need more mitochondria. **1**

_____

_____

2. Ions move in and out of cells by passive and active processes.

The table shows the concentration of three ions outside and inside a human cell.

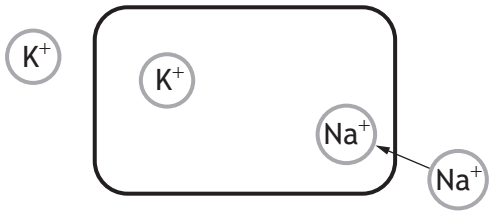
Ions	Outside cell (mM)	Inside cell (mM)
Sodium (Na^+)	145	12
Potassium (K^+)	4	139
Chloride (Cl^-)	116	4

(a) Name the process by which sodium ions move into this cell. **1**

_____

(b) (i) Potassium ions (K^+) move by active transport. **1**

Using the information in the table, complete the diagram, by drawing an arrow, to show the direction in which the potassium ions move.



(ii) Name the type of molecule, found in a cell membrane, which is involved in moving ions by active transport. **1**

_____

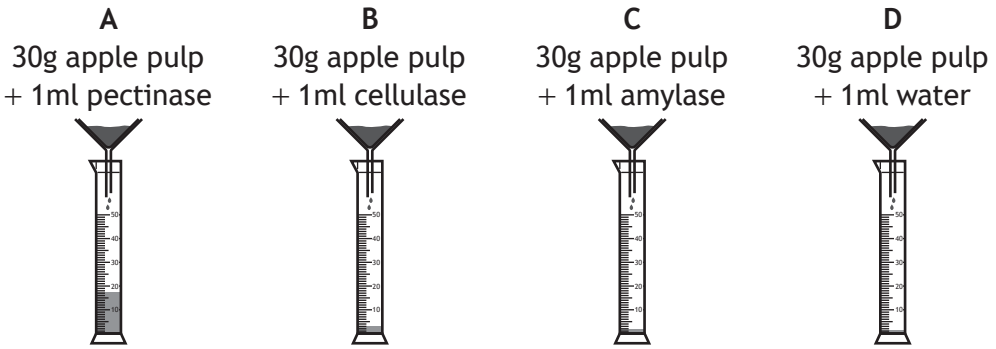
(c) Calculate how many times greater the concentration of chloride ions is outside the cell compared to inside the cell. **1**

Space for calculation

_____ times greater

3. Enzymes are used to help extract juice from fruit pulp.

The diagrams show a student's investigation into the effectiveness of different enzymes on the volume of juice produced.



After 30 minutes, the volume of apple juice collected was measured and the procedure was then repeated.

The average for each cylinder was calculated and the results are shown in the table.

Cylinder	Liquid added to apple pulp	Average volume of apple juice collected (ml)
A	pectinase	17·6
B	cellulase	3·2
C	amylase	1·8
D	water	1·6

(a) Name the enzyme which was the least effective at extracting juice from the apple pulp. **1**

_____

(b) Enzymes can be involved in two types of reaction. The extraction of juice from apple pulp is an example of a degradation reaction.

Name the other type of reaction. **1**

_____

[Turn over

3. (continued)

(c) Cellulase is an enzyme that speeds up the breakdown of a component of the plant cell wall.

Name this component. **1**

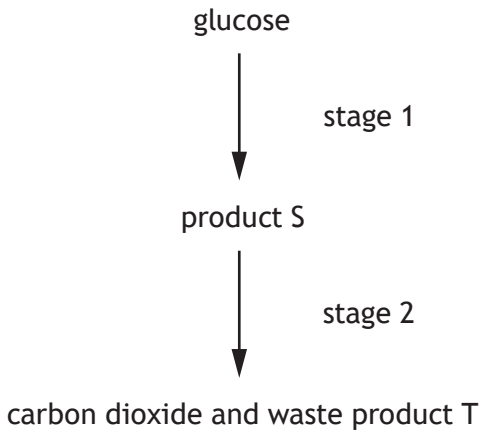
_____

(d) All of the enzymes broke down at least some part of the apple pulp to produce apple juice.

What name is given to the substance that an enzyme breaks down? **1**

_____

4. The diagram represents the process of aerobic respiration in yeast cells.



(a) (i) Name product S and waste product T. **2**

S _____

T _____

(ii) Name the substance that must be present for yeast to respire aerobically. **1**

_____

(iii) Identify which of the two stages releases the larger quantity of ATP. **1**

Stage _____

[Turn over

MARKS

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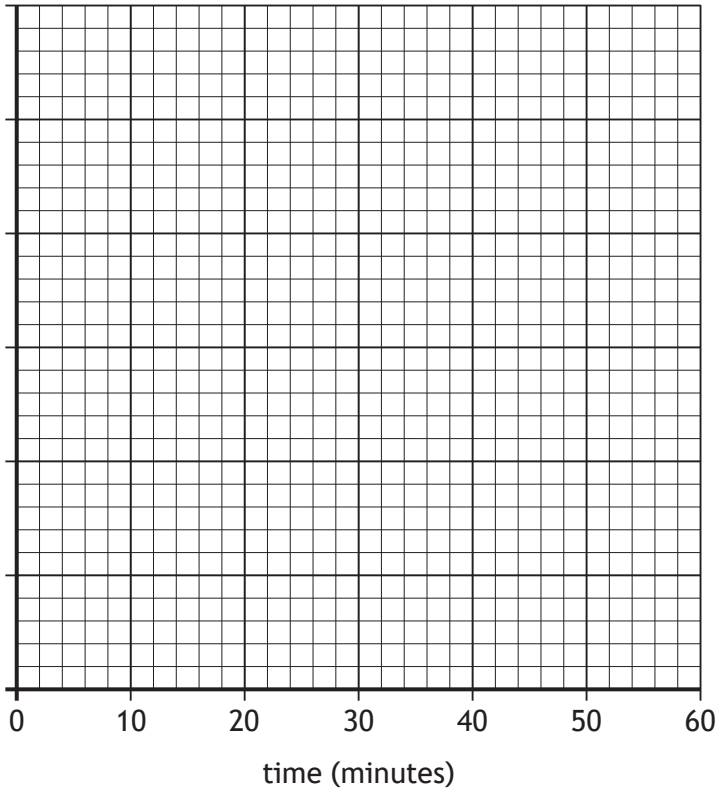
(b) An investigation was carried out into the respiration of yeast. A dough was made containing live yeast and left in optimum conditions. As the yeast respired, the carbon dioxide produced caused the dough to rise. The volume of the dough was measured every 10 minutes for 60 minutes.

The results are shown in the table.

Time (minutes)	Volume of dough (cm^3)
10	8
20	14
30	22
40	26
50	28
60	28

(i) On the grid, complete the vertical axis and plot a line graph to show the effect of time on the volume of dough. **2**

(An additional grid, if required, can be found on *page 31*)



4. (b) (continued)

MARKS DO NOT WRITE IN THIS MARGIN

(ii) Predict the volume of dough at 50 minutes if the experiment was carried out at a lower temperature. **1**

Give a reason for your answer. **1**

Prediction _____ cm^3

Reason _____

_____

[Turn over

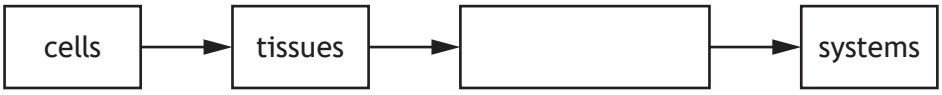
5. New cells are produced by mitosis throughout life.

(a) Name the cells, found in the early stages of an embryo's development, which have the potential to develop into specialised cells. **1**

_____

(b) As cells become specialised they are organised according to their function.

Complete the flowchart to show the levels of organisation found within a multicellular organism. **1**



(c) State a reason, other than growth, why cells continue to be produced throughout life. **1**

_____

(d) Lymphocytes and phagocytes are specialised white blood cells.

Describe the different ways in which these two types of cell destroy pathogens. **2**

_____

_____

_____

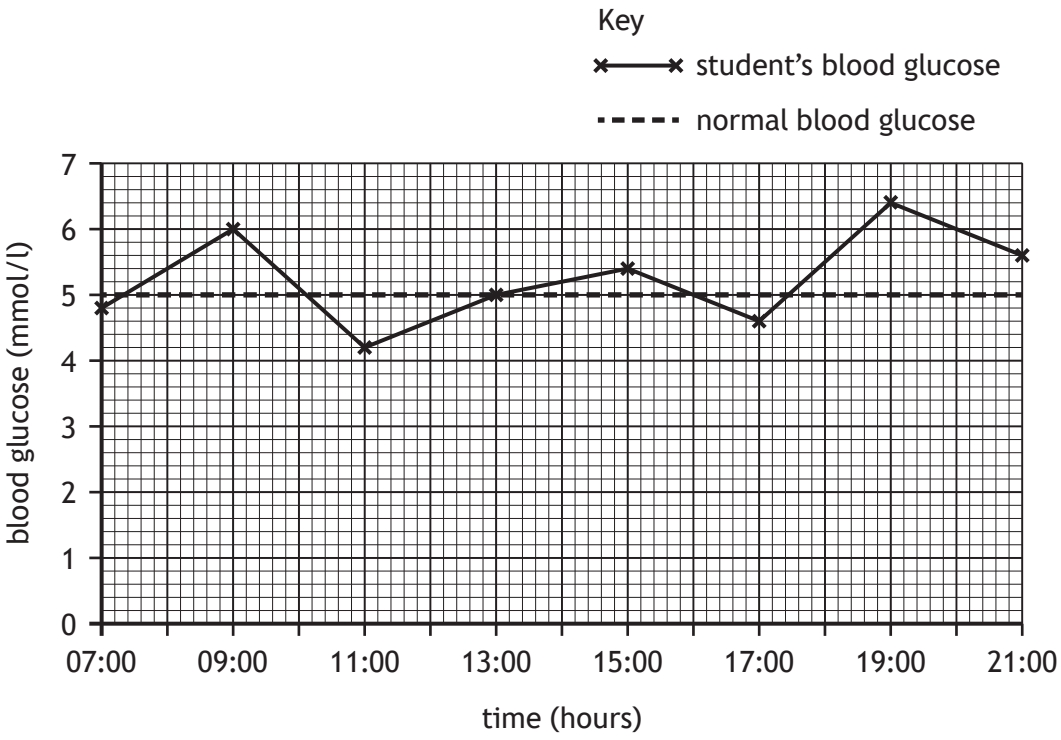
_____

(e) Name the system in the human body which destroys pathogens. **1**

_____

6. As part of a study into the health of a group of students, blood glucose readings were taken over a period of time.

The graph shows the readings for one of the students.



(a) Identify the time when this student's blood glucose reached its maximum value. **1**

_____ hours

(b) Calculate the percentage decrease in blood glucose between 09:00 and 11:00 hours. **1**

Space for calculation

_____ %

[Turn over

6. (continued)

(c) By 13:00 hours the blood glucose level had returned to normal from its lowest point.

Describe how blood glucose is returned from low to normal levels in the human body. **4**

7. An experiment was carried out to investigate stem height in pea plants.

The parental phenotypes were tall and dwarf as shown in the diagrams.



tall dwarf

stem height

(a) The parent plants were both homozygous. When they were crossed the F_1 generation were all tall. These plants were then crossed with each other to produce the F_2 generation.

(i) Explain what is meant by the term homozygous. **1**

_____

_____

(ii) The dwarf characteristic is recessive.

Using the information given, explain how this is known. **1**

_____

_____

[Turn over

7. (continued)

(b) (i) The expected ratio in the F_2 generation was 3 tall to 1 dwarf.

Calculate the expected number of tall plants if there were 144 plants produced in this generation. **1**

Space for calculation

_____ tall plants

(ii) The results obtained in the F_2 generation differed from the expected results.

The actual results were 90 tall and 36 dwarf plants.

Calculate the simplest whole number ratio for these results. **1**

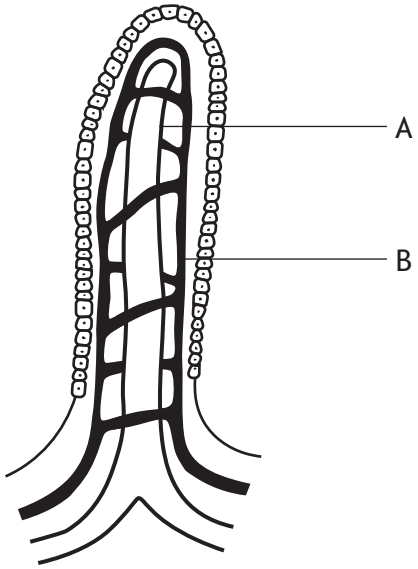
Space for calculation

_____ : _____

tall dwarf

8. Nutrients from food are absorbed into villi found in the small intestine.

The diagram represents a single villus.



(a) Name one type of molecule absorbed by each of the labelled structures. **2**

A _____

B _____

(b) Explain why having a large number of villi improves the efficiency of absorption in the small intestine. **1**

_____

_____

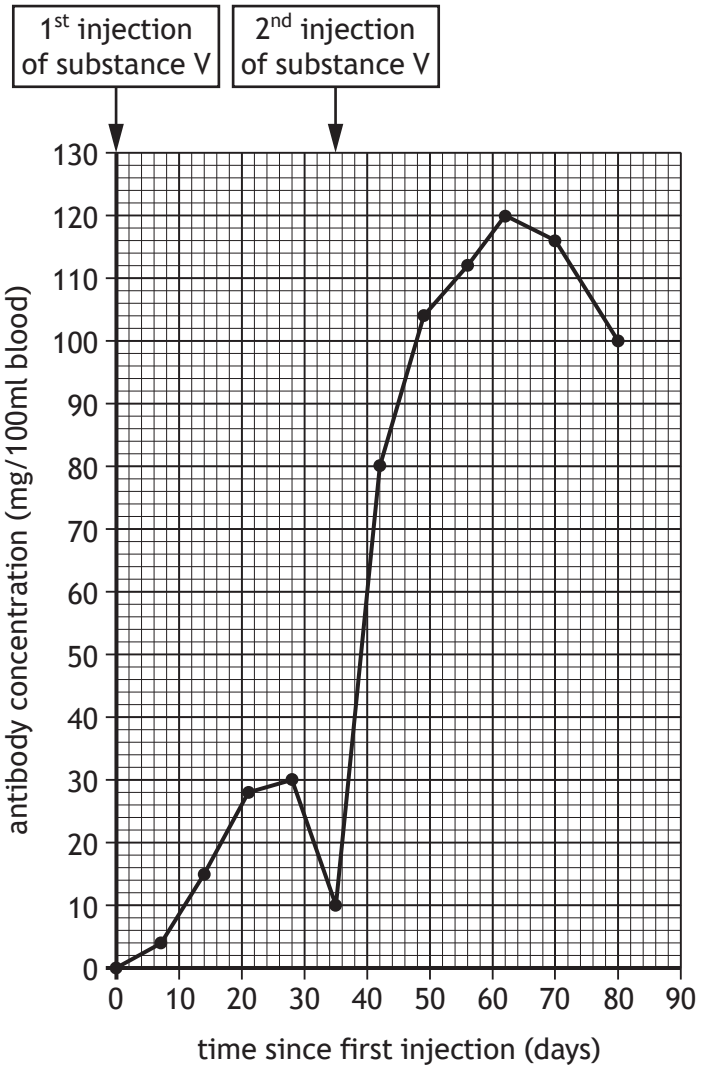
(c) Identify the structural feature of a single villus, that is also found in an alveolus, which increases the efficiency of absorption. **1**

_____

[Turn over

9. In a study, volunteers were injected with substance V on day 0 and again on day 35. Substance V causes antibodies to be produced.

The graph shows the antibody concentration in the blood of one of the volunteers in response to the two injections.



(a) Identify the maximum antibody concentration following the first injection, but before the second. **1**

_____ mg/100 ml

(b) Calculate the number of days after the second injection that it took for the antibody concentration to reach its maximum value. **1**

Space for calculation

_____ days

9. (continued)

(c) The second injection caused a higher concentration of antibody to be produced than the first.

Give **two** other differences in the antibody production in response to the two injections. **2**

1 _____

_____

2 _____

_____

(d) If the trend continues as shown in the graph, predict the antibody concentration on day 90. **1**

Space for calculation

_____ mg/100 ml

[Turn over

10. A study found that exercise can reduce the risk of developing some types of cancer.

Researchers examined the level of exercise of 1·4 million people over an 11 year period and recorded any diagnosis of cancer and when it occurred.

Participants were asked to keep their own record of the frequency and intensity of their exercise.

Those participants who exercised were found to have a reduced risk of developing some types of cancer. These results are shown in the table.

Type of cancer	Average risk reduction (%)
lung	26
kidney	23
stomach	22
myeloma	17
bladder	13

The study did not take into account factors such as diet and smoking, which may have affected the results. The fact that participants were asked to record their own exercise is another limitation of this study.

The doctor in charge of the research suggested that these results support the promotion of exercise as a means of reducing the risk of cancer.

However, it was suggested by a different researcher that further studies would need to be carried out before the results could be considered valid.

(a) Most research starts off with a question. For example, 'Is enzyme activity affected by pH?'

Suggest a question that could have led to the research described above. **1**

_____

_____

(b) Name the type of graph that should be used to present the results shown in the table. **1**

_____

10. (continued)

(c) Suggest one reason why the participants recording their own exercise is described as a limitation. **1**

(d) Three factors that could affect the results are age, duration of exercise and type of exercise.

Choose one of these factors.

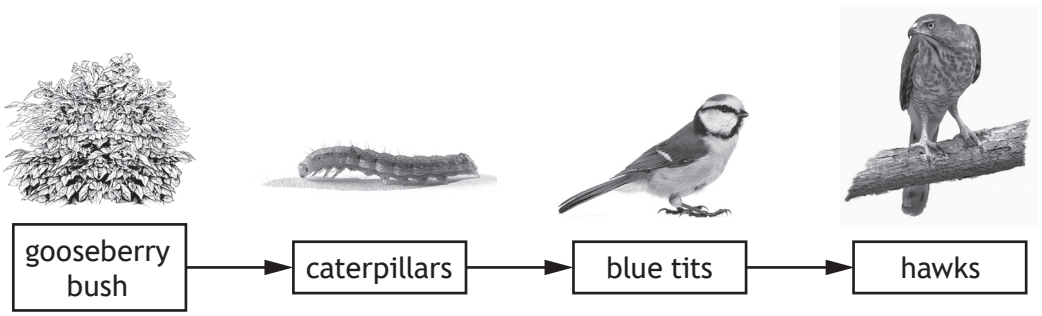
Describe how the study would be carried out to take the chosen factor into account and improve the validity of the results. **1**

Factor

Description

[Turn over

11. (a) The food chain represents the transfer of energy from organism to organism.



(i) Name the process carried out by the gooseberry bush that allows energy to enter the food chain. **1**

_____

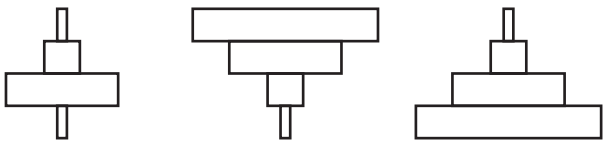
(ii) The blue tits are consumers.

Give another ecological term that describes the role of the blue tits in this food chain. **1**

_____

(iii) A food chain can be shown as a pyramid of numbers.

pyramid A pyramid B pyramid C



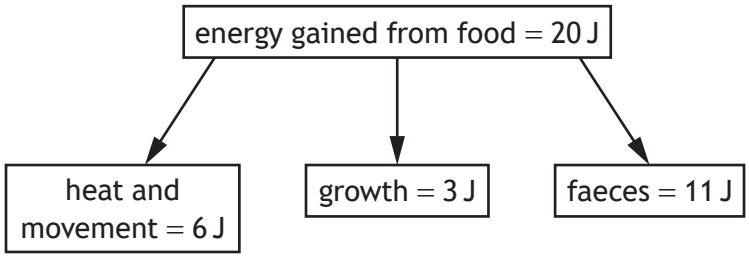
Identify which pyramid would be used to represent the food chain shown above. **1**

Pyramid _____

11. (continued)

(b) The diagram shows the fates of 20 joules (J) of energy gained by a caterpillar from food.





(i) Calculate the percentage of the caterpillar's energy intake that is used for growth. **1**

Space for calculation

_____ %

(ii) Tick one box to identify which of the fates allows energy to be passed onto the next level in the food chain shown. **1**

☐ Heat and movement ☐ Growth ☐ Faeces

[Turn over

12. Students carried out an investigation into the effect of soil moisture on the percentage ground cover of moss in their school lawn.

A quadrat was used to estimate the percentage ground cover of moss at several sites. Soil moisture was also measured at each of the sites.

The results of the investigation are shown in the table.

Sample site	Average soil moisture (%)	Ground cover of moss (%)
1	18	86
2	14	70
3	15	80
4	11	58
5	13	65
6	12	60
7	22	98
8	30	99
9	35	100

(a) Describe the relationship between average soil moisture and percentage ground cover of moss. **1**

(b) Describe how the reliability of these results could be increased. **1**

12. (continued)

(c) The students also used a soil thermometer to gather data on soil temperature at each of the sample sites.

(i) Describe what the students should have done each time they used the soil thermometer, to ensure valid results. **1**

_____

_____

(ii) Soil moisture and temperature are examples of abiotic factors.

Name **one** other abiotic factor that can affect plant growth. **1**

_____

(d) (i) The students observed a species of lichen growing on some of the damper parts of the lawn.

Lichens are indicator species.

State what is meant by the term 'indicator species'. **1**

_____

_____

(ii) Identify the sample site that would be least likely to contain this lichen. **1**

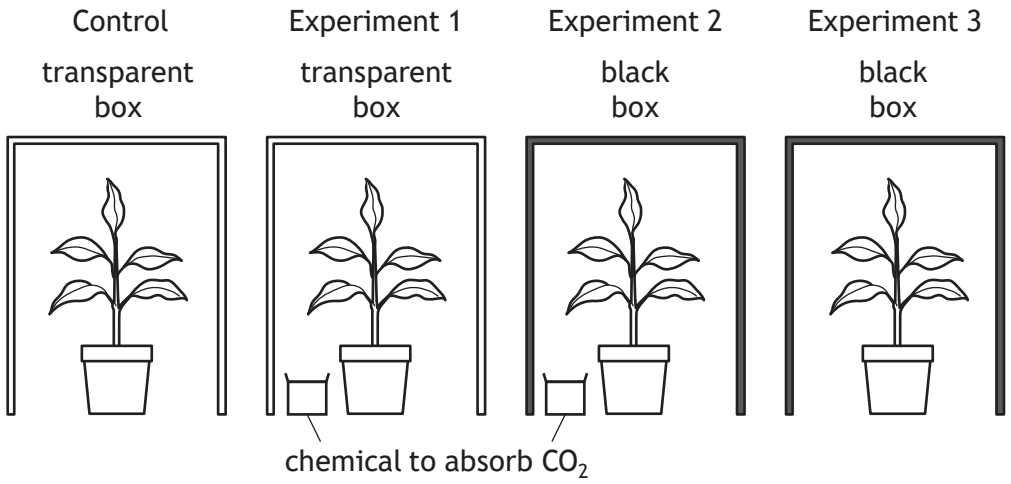
Sample site number _____

[Turn over

13. An investigation was carried out into the conditions needed for photosynthesis.

Before starting the investigation, four plants were de-starched by placing them in the dark for 24 hours.

The plants were then placed in the conditions shown and then left for 48 hours.



The leaves of each plant were then tested for the presence of starch.

(a) (i) Tick the box(es) to show in which of the following starch would be present after 48 hours. **1**

☐ Control ☐ Experiment 1

☐ Experiment 2 ☐ Experiment 3

(ii) Explain the purpose of the control. **1**

_____

_____

(iii) Explain why experiment 2 is invalid. **1**

_____

_____

13. (a) (continued)

(iv) State the factor that is being investigated in experiment 3. **1**

_____

(b) Name the product of carbon fixation that is converted to starch. **1**

_____

[Turn over

14. Antibiotic drugs are only effective in the treatment of bacterial infections. They either kill or prevent the growth of bacteria. Sometimes antibiotics are given but do not have any effect on the particular infection.

(a) The table gives information about the number of cases treated with antibiotics and the success rate for a range of infections.

Type of infection	Number of these cases treated with antibiotics (millions)	Number of these cases successfully treated with antibiotics (millions)	Success rate (%)
ear	23	16·1	70
common cold	18	0	0
chest	16	3·2	20
throat	15		48

(i) Complete the table by calculating the number of cases of throat infections successfully treated with antibiotics. **1**

Space for calculation

(ii) Antibiotics have no effect in the treatment of the common cold.

From the information given, suggest a reason why antibiotics have no effect on the pathogen that causes the common cold. **1**

_____

_____

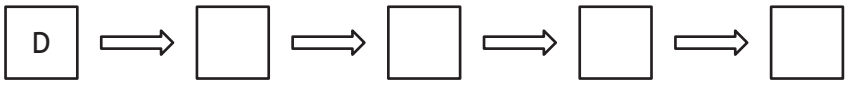
14. (continued)

(b) The overuse of antibiotics has caused populations of antibiotic-resistant bacteria to evolve. The stages in their evolution are described in the table.

Stage	Description
A	Random mutations in bacteria make some of the bacteria antibiotic-resistant
B	The next generation of bacteria are antibiotic-resistant
C	The surviving bacteria reproduce and pass on the alleles for antibiotic resistance to their offspring
D	Bacterial populations show genetic variation
E	When antibiotics are used, the non-resistant bacteria are killed and the resistant bacteria survive

(i) Place letters from the table in the boxes, to show the order of the stages describing how populations of bacteria become resistant to antibiotics. **1**

The first stage has been completed.



(ii) Name the process by which the bacteria, best adapted to their environment, survive and reproduce. **1**

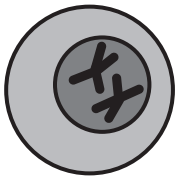
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(iii) Name an environmental factor that can increase the rate of mutation. **1**

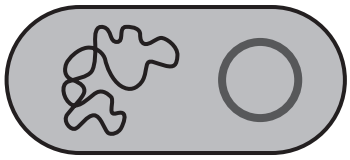
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[Turn over for next question

15. Scientists use bacteria to act as host cells, to produce enzymes commercially by the process of genetic engineering.



source cell



bacterial cell

(a) Name the structure in the source cell from which the required gene is extracted during the first stage of this process. **1**

_____

(b) Describe the stages of the process that would be used to produce genetically engineered bacteria, after the required gene has been extracted from the source cell. **4**

_____

_____

_____

_____

_____

_____

_____

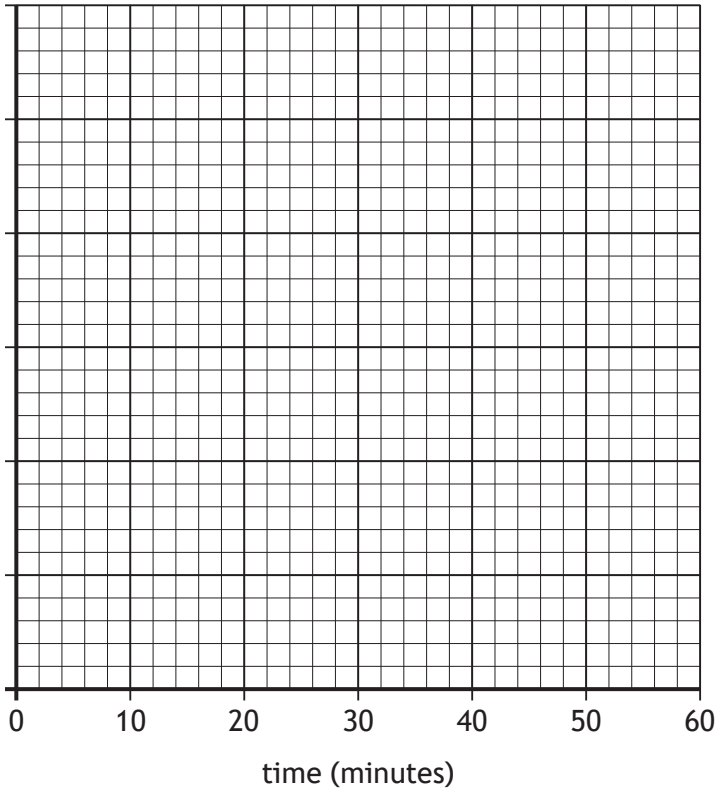
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[END OF QUESTION PAPER]

ADDITIONAL SPACE FOR ANSWERS AND ROUGH WORK

Additional grid for question 4 (b) (i)



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